

Models and Theories

What are they?

A Theory

A scientific *theory* is a set of principles or a framework that explains some phenomena, and that is:

(i) **Testable**

- ▶ Otherwise it is not scientific. Karl Popper says a theory should be *falsifiable*.

(ii) **Economical, or parsimonious**

- ▶ A theory should be in some sense simpler than the phenomena it explains, or it is merely a description.

(iii) **General**

- ▶ A theory should be able to explain a class of phenomena, not just a single instance.

Examples: Darwinian evolution, Relativity, Quantum Mechanics, Plate Tectonics

- Ideally based on some a priori foundation, at some level.
- Theories are subsets of models: A theory is a model, but not all models are theories.
- No need for a theory to be based on equations.